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MILITARY SPECIFICATION

POWER UNIT TEST SET

(For Torpedo Mk 46 Mod 1)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 Scope. This specification covers the requirements for the procurement of one type of power unit test set (see 6.1.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

DRAWINGS

NAVAL ORDNANCE SYSTEMS COMMAND

DL 2539015	Power Unit and all documents, listed thereon
DL 2540730	Power Unit Test Set, and all documents listed thereon

PUBLICATIONS

NAVAL ORDNANCE SYSTEMS COMMAND

WD 28933	Special Manufacturing Procedure for Calibration of Power Unit Test Set, DL 2540730

(Copies of drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity as directed by the contracting officer.)

FSC 1355

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3. REQUIREMENTS

3.1 Basic requirements

3.1.1 Contractor-furnished data. No data is required by this specification or by applicable documents referenced in Section 2, unless specified in the contract or order (see 6.2).

3.1.2 Materials and treatments of materials and parts. Unless otherwise specified in the contract or order, all materials and treatments of materials and parts used in the manufacture of the power unit test set shall be in accordance with the requirements specified herein and with the applicable documents listed in DL 2540730.

3.1.3 Requests for deviation. Each request that a deviation be authorized shall have been submitted to the Procuring Activity, and the deviation shall have been authorized by the Naval Ordnance Systems Command or by a Naval Activity delegated such authority, prior to the production of the affected item, or prior to the performance of a particular operation on the item. The deviation shall apply only to this specific item or operation. The request shall define the suggested deviation in sufficient detail to permit the Procuring Activity to evaluate its effect on performance and reliability, and the reasons for the suggested deviation shall be stated. Each request that a deviation be authorized, and the authorization, shall state specifically the extent of the departure, the identifying number of the item, the contractor (or bidder), and the contract number. A deviation is essentially temporary in character, and usually represents a departure limited to a specific period, to a specific number of items or to the contract to which it applies. It may require an equitable reduction in contract price if a decrease in the cost of performance results therefrom.

3.2 Performance and product characteristics

3.2.1 Input power. The power unit test set (hereafter referred to as test set) shall provide means for switching power on and off to the test set and to the power unit being tested. Indicator lamps shall be provided to ascertain that power is being applied. Maximum and minimum load conditions shall be provided for the power unit, a panel meter and test points shall be provided for measuring voltage levels. The input power requirements of the test set shall be as specified in Table I.

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Table I

Input Power

Voltage	Current
115 ±10 Volts alternating current (v ac)	15 amperes maximum (amps max)

3.2.2 Test set outputs. The test set shall provide two voltage adjustable outputs one of +40 ±5 vdc and one of -40 ±5 vdc at 7 amperes for evaluating the performance of the power unit. Roll, yaw, and pitch drive commands shall be provided to simulate autopilot outputs.

3.2.3 Performance requirements. The test set shall provide the following test functions:

3.2.3.1 Power supply load. A meter and test points for monitoring power supply output voltages.

3.2.3.2 Exploder arm load. A switch to apply exploder arm load manually to power supply. A test point to monitor exploder arm voltage.

3.2.3.3 Shunt field load. A (momentary) switch to apply shunt field load manually to power supply. Test points to monitor shunt field voltage.

3.2.3.4 4.8 KC oscillator load. A 4.8 kc oscillator load and test points for measuring voltage and phase shift of 4.8 kc signal from power supply.

3.2.3.5 Actuator controller load. Switches for controlling the autopilot drive signals to the actuator controller. Simulated motor loads and indicator lamps to show load conditions. A pulsed load simulates the normal mode of motor operation.

3.3 Workmanship. The workmanship and finish shall be of sufficiently high grade to assure satisfactory operation consistent with the requirements of this specification. All parts used in the test set shall be thoroughly clean and free of excess materials, chips, and loose or spattered foreign materials. The standards of workmanship, subject to any qualifications stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

3.4 Identification. The test set shall be permanently identified in accordance with the applicable documents listed in DL 2540730.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspections of the test set shall be classified as follows:

4.2.1 Quality conformance inspection

(a) Quality conformance sampling (see 4.3.1)

(b) Quality conformance tests and examinations (see 4.5)

4.3 Sampling

4.3.1 Quality conformance sampling. The tests and examinations specified in 4.5 shall be conducted on each test set to be delivered; therefore, there shall be no sampling plan for acceptance. Defective units shall be rejected.

4.4 Test conditions. Unless otherwise specified, the test set shall be subjected to tests under the following conditions:

(a)	Temperature	Room ambient (60 to 95 degrees F)
(b)	Humidity (Relative)	Room ambient, up to 95 percent
(c)	Altitute	Normal ground atmospheric pressure

4.5 Quality conformance tests and examinations. Unless otherwise specified each test set delivered under contract shall be subjected to the tests and examinations contained in 4.5.3.

4.5.1 Special notes

4.5.1.1 Test sequence. Unless otherwise noted, the sequence of testing procedures presented herein is mandatory.

4.5.1.2 Test results. Where space is provided in the Appendix Report Form to indicate the value or specific reading obtained, the specific

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reading shall be recorded if a limit or limits is given in the Limit column for the test. If limits are not stated in the Limit column, the individual performing the tests or inspection shall initial in the space provided in the reading column indicating that the requirements were met satisfactorily.

4.5.1.3 Quality conformance criteria. The failure of the end product to pass any examination or test of this section will automatically classify the unit as nonconforming. When nonconforming units are corrected by the contractor, the unit shall be reinspected. The reinspection of a nonconforming unit may be limited to the test or examination which defined the nonconformance, or when so directed by the cognizant inspector, a complete re-examination and retest of the unit may be required. Nonconforming units, which have not been corrected, will be considered for acceptance only upon formal application by the contractor to the Naval Ordnance Systems Command via the cognizant Government representative.

4.5.2 Quality conformance test equipment. The items of test equipment shown in Table II are required to perform the quality conformance tests specified herein. All items of test equipment shall be of sufficient accuracy and precision to permit valid performance of the required tests.

Test instruments shall be calibrated at regular intervals and the procedures for and the results of all calibrations shall be subject to approval of the cognizant Government Inspector. Items having equivalent characteristics may be substituted where necessary. Verify that all test equipment used is identified as being within calibration.

Table II

Test Equipment

Item	Identification
Resistor (3) 1000 ohms, 3 watts, 5%	Commercial
AC Current Probe Sensitivity 1.0 millivolt per milliampere ±1% at 1.0 kilocycle; Response ±2% 100 cps to 3.0 mc/s; Rise time less than 20 usec.	Hewlett-Packard model 456A
Resistance Bridge Range 0.001 to 9,999,000 ohms; Accuracy ±0.15% +0.01 ohm for 1 to 40 K	Leeds & Northrup No. 5305 Test Set

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Table II (Cont'd)

Test Equipment

Item	Identification
Oscilloscope Bandpass from dc to 15 megacycles; Rise time from 0.023 microsecond; Accuracy 3% from 0.05 to 20 volts/cm.	Tektronix Model 535A with type CA preamplifier
Digital Voltmeter Range: 0.001 to 999.9 vdc or vrms; DC accuracy: ±0.05% of reading ±1 digit; AC accuracy: ±0.1% of reading ±2 counts: Input impedance: 10 megohms, 20 pf.	Hewlett-Packard Model 3440A with 3445A AC/DC Range Unit
Multimeter Resistance range 0 to 20 megohms; Current range 0 to 10 amperes; Voltage range 0 to 5000 volts ac or dc; Accuracy ±3% full scale for dc, ±5% full scale for ac.	Simpson Model 260
Digital Voltmeter Range: 0.001 to 999.9 vdc or vrms; DC accuracy: ±0.05% of reading ±1 digit; AC accuracy: ±0.1% of reading ±2 counts: Input impedance: 10 megohms, 20 pf.	Hewlett-Packard Model 3440A with 3445A AC/DC Range Unit
4.8 KC/s Power Oscillator Freq. stability less than ±0.25% Distortion less than 0.5%; P_o = 20 watts; E_o = 0 to 120 volts--800 ohms load, 0 to 60 volts--200 ohms load, 0 to 28 volts-- 40 ohms load.	Industrial Test Equipment Model 1500 -- 4.8 kc/s

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4.5.3 Test procedure

4.5.3.1 Continuity test

4.5.3.1.1 Verify that all power switches on test set are OFF.

CAUTION

Do not connect the test set ac power cable.

4.5.3.1.2 Install connectors P2B and P101B to corresponding connectors located on side of test set. Using resistance bridge, check resistance or continuity between pins of connector P1501 and corresponding test point on power supply test panel as indicated below. Record results.

P1501 Pin	Power Supply Test Panel Test Point
A	EXPLODER ARM
B	+90 V (UPPER)
C	+90 V (LOWER)
D	SPARE TP117
E	-30 VG
F	GRD
H	4.8KC GRD
J	4.8KC GRD
N	+15 V
P	-15 V
R	+30 V
S	-30 V
UU	Neg. (-) terminal of +40 V Reg. Power Supply
VV	Pos. (+) terminal of -40 V Reg. Power Supply

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P1501 Pin	Power Supply Test Panel Test Point
T	+5 V
U	COMMON (ACTUATOR CONTROLLER TEST PANEL)
V	4.8KC IN ∅
X	4.8KC GRD
Y	400 CPS GRD
CC	SPARE (TP119)
LL	SPARE (TP118)
WW	+40 V
XX	-40 V
RR	SHUNT FIELD (+)
SS	SHUNT FIELD (-)

4.5.3.1.3 Set POWER SUPPLY TEST PANEL POWER switch to ON and continue continuity test.

P1501 Pin	Pin
WW	+40 VDC POWER SUPPLY OUTPUT TERMINAL
XX	-40 VDC POWER SUPPLY OUTPUT TERMINAL

4.5.3.1.4 Set POWER SUPPLY TEST PANEL POWER switch to OFF.

4.5.3.1.5 Position power supply test panel controls as indicated below. Check resistance or continuity using resistance bridge between following connector pins and test points located on power supply test panel. Use multimeter (function +dc) for EXPLODER ARM OUT measurements only. Record results.

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Control	Position	P1501 Pin	Test Point
COMPUTER ENABLE	OFF	L	GRD
COMPUTER ENABLE	ON	L	P1501 Pin P
EXPLODER ARM	OUT	A (multimeter - lead)	GRD (multimeter + lead)
EXPLODER ARM	OUT	A (multimeter + lead)	GRD (multimeter - lead)
EXPLODER ARM	IN	A	+40 V

4.5.3.1.6 Position SHUNT FIELD switch as indicated below. Check resistance or continuity using resistance bridge between following pins and test points located on power supply test panel. Record results.

SHUNT FIELD Switch Position	P1501 Pin	P1501 Pin
OUT	RR	SS
IN	RR	SS

4.5.3.1.7 Check resistance or continuity between pins of connector P1501 and corresponding test point on actuator controller test panel as indicated below. Record results.

P1501 Pin	Actuator Controller Test Panel Test Point
M	INSTRUMENT 1
VV	COMMON
W	INSTRUMENT 2
Z	PITCH
AA	YAW
BB	ROLL
DD	RIGHT UPPER RUDDER
EE	LEFT UPPER RUDDER

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P1501 Pin	Actuator Controller Test Panel Test Point
HH	-40 V
JJ	UP ELEVATOR
KK	DOWN ELEVATOR
MM	RIGHT LOWER RUDDER
NN	LEFT LOWER RUDDER
PP	PITCH INHIBIT
TT	+40 V
FF	4.8KC GRD (POWER SUPPLY TEST PANEL)
K	+40 V
COMMON (TP11)	COMMON (TP14)

P1501 Pin	Pin
HH multimeter (+)	Multimeter (-) -40 VDC POWER SUPPLY output jack. (Set the ACTUATOR CONTROLLER TEST PANEL POWER switch to ON.)
TT multimeter (-)	Multimeter (+) +40 VDC POWER SUPPLY output jack.
U	Power Supply Common output jack. (Set the ACTUATOR CONTROLLER TEST PANEL POWER switch to OFF.)

4.5.3.2 Power supply test panel load resistance test

4.5.3.2.1 Check resistance using resistance bridge between the following pins of P1501. Set POWER PANEL circuit breaker to ON. Using digital voltmeter, adjust plus and minus 40v regulated power supplies to 39.8 to 40.2 vdc. Set POWER SUPPLY TEST PANEL POWER switch to ON. The LOAD switch should be positioned as indicated when making resistance measurement.

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4.5.3.2.2

P1501 Pins	Load Switch Position	Resistance
T to U	MIN	234 to 246 ohms
T to U	MAX	69.0 to 72.0 ohms
N to U	MIN	86.5 to 89.5 ohms
N to U	MAX	48.0 to 50.0 ohms
R to U	MIN	247 to 253 ohms
R to U	MAX	127 to 132 ohms
P to U	MIN	15.0 to 17.0 ohms
P to U	MAX	13.0 to 14.5 ohms
E to U	MIN	80.5 to 83.5 ohms
E to U	MAX	56.5 to 59.5 ohms
S to U	MIN	158 to 162 ohms
S to U	MAX	122 to 125 ohms

Set POWER SUPPLY TEST PANEL POWER switch to OFF position.

4.5.3.2.3 Meter switch test

CAUTION

Disconnect J101B from P101B on power unit test set.

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Connect plus and minus 40-volt regulated power supplies as indicated below to obtain specified deflections of meter on test set:

POWER SUPPLY TEST PANEL TEST POINT		POWER SUPPLY TEST PANEL TEST POINT		Meter Switch Position	Deflection
+40V	+40 vdc Regulated power supply output	4.8KC GRD	+40 vdc Regulated supply common	+40	Upscale ≈40 vdc
+30V	+40 vdc Regulated power supply output	4.8KC GRD	+40 vdc Regulated supply common	+30	Upscale ≈40 vdc
+15V	+40 vdc Regulated power supply output	4.8KC GRD	+40 vdc Regulated supply common	+15	Upscale ≈40 vdc
+5V	+40 vdc Regulated power supply output	4.8KC GRD	+40 vdc Regulated supply common	+5	Upscale ≈40 vdc
-40V	-40 vdc Regulated power supply output	4.8KC GRD	+40 vdc Regulated supply common	-40	Upscale ≈40 vdc
-30V	-40 vdc Regulated power supply output	4.8KC GRD	+40 vdc Regulated supply common	-30	Upscale ≈40 vdc
-15V	-40 vdc Regulated power supply output	4.8KC GRD	+40 vdc Regulated supply common	-15	Upscale ≈40 vdc
-30VG	-40 vdc Regulated power supply output	4.8KC GRD	+40 vdc Regulated supply common	-30G	Upscale ≈40 vdc

4.5.3.3 4.8KC oscillator load test

NOTE

All test instruments shall be isolated from ground when performing 4.5.3.3.1 through 4.5.3.3.7.

4.5.3.3.1 Connect P101B to J101B on power unit test set. Set POWER switches on actuator controller and power supply test panels in OFF position. Connect 4.8 kc oscillator as shown in Figure 1(a). Oscillator frequency shall be 4800 ±10 cps.

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4.5.3.3.2 Connect digital voltmeter to 4.8 kc in ∅ and 4.8 kc GRD test points on power supply test panel. Position LOAD switch to MIN. Set 4.8 kc oscillator to 25.9 to 26.1 volts rms output. Disconnect digital voltmeter.

4.5.3.3.3 Connect digital voltmeter and ac current probe as shown in Figure 1(b). Digital voltmeter shall indicate 280 ±15 millivolts rms. Record voltage indication.

4.5.3.3.4 Position LOAD switch to MAX position. Set 4.8 kc oscillator to 25.9 to 26.1 volts rms output. Digital voltmeter shall indicate 410 ±15 millivolts rms. Record voltage indication.

4.5.3.3.5 Position LOAD switch to MIN. Position oscilloscope controls as follows:

Control	Position
HORIZONTAL DISPLAY	A
TRIGGER MODE	AUTO
TRIGGER SLOPE	EXT +
VARIABLE (TIME/CM)	CALIBRATED
TIME/CM	20 usec
MODE	ALTERNATE
AC/DC (CHANNELS A and B)	AC
POLARITY (CHANNEL A and B)	NORMAL

4.5.3.3.6 Adjust both channels A and B trace vertical positions to center of graticule. Remove ac current probe from digital voltmeter. Connect oscilloscope and ac current probe as shown in Figure 1(c).

NOTE

The current probe direction arrow must point toward P1501-V.

Set TRIGGER MODE to AC (LF REJECT). Set MODE switch to B ONLY. Using TRIGGER LEVEL, STABILITY and HORIZONTAL POSITION controls, adjust trace to trigger when it crosses zero at left-hand edge of graticule. Set MODE switch to ALTERNATE.

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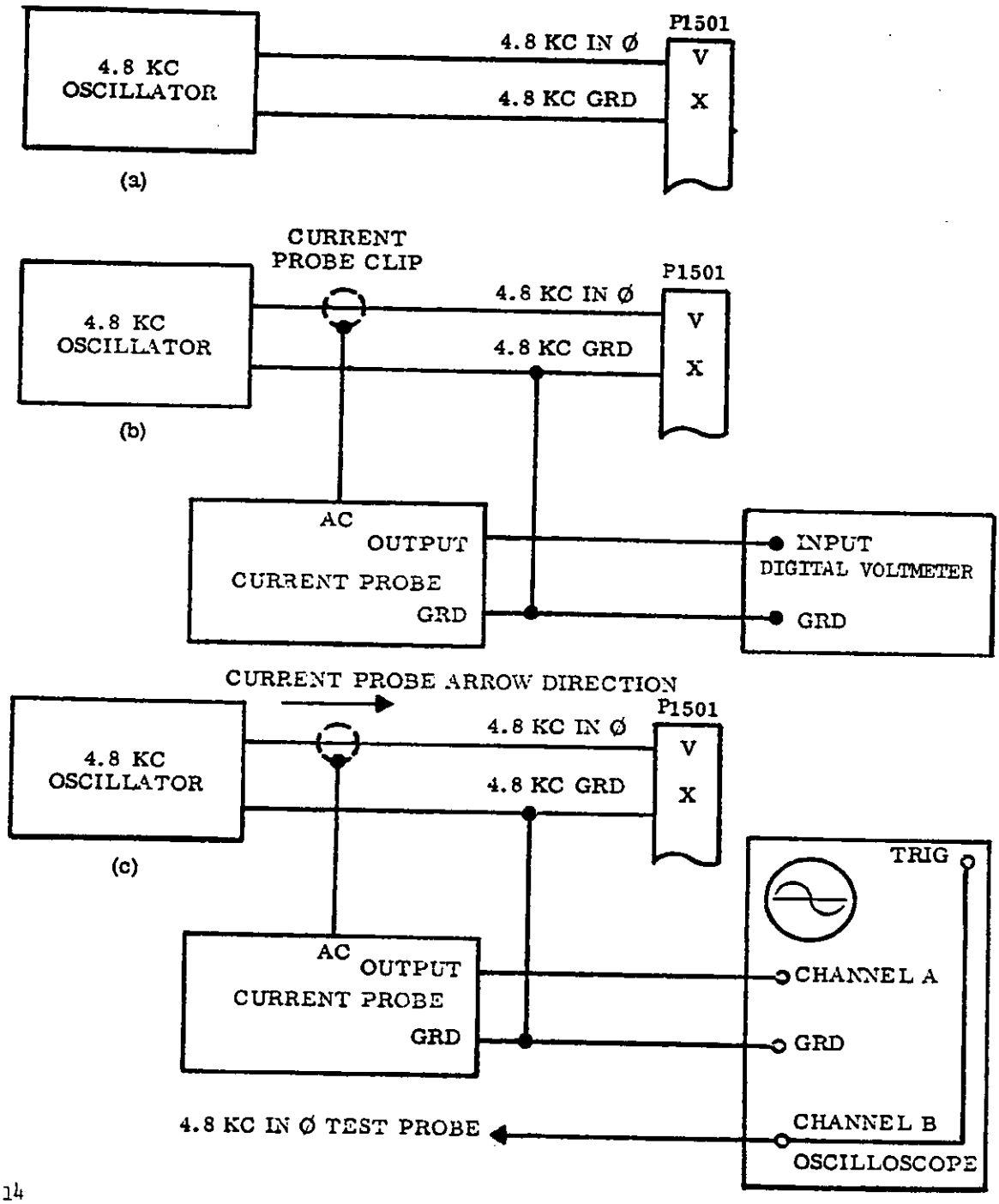


Figure 1. 4.8KC Oscillator Load Test

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4.5.3.3.7 Using VARIABLE VOLTS/CM controls, adjust both channels for equal full scale amplitudes. Set TIME/CM switch to 10 usec. Adjust VARIABLE TIME/CM controls so that 180° of the CHANNEL B trace covers 10 centimeters. Using HORIZONTAL POSITION control, position negative-going axis crossing to center of graticule. Turn 5 X magnifier. The difference in the zero axis crossing shall be 8.9 to 10.0 centimeters. Record phase shift value.

4.5.3.3.8 Set LOAD switch to MAX. Turn off 5 X magnifier. Set TIME/CM switch to 20 usec. Using VARIABLE VOLTS/CM controls, adjust both channels for equal full scale amplitude while keeping channel B trace triggered as it crosses zero at left-hand edge of graticule. Set TIME/CM switch to 10 usec. Adjust VARIABLE TIME/CM controls so that 180° of CHANNEL B trace covers 10 centimeters.

4.5.3.3.9 Using HORIZONTAL POSITION control, position negative-going axis crossing to center of graticule. Turn on 5 X magnifier. The difference in the zero axis crossing points shall be 3.0 to 4.0 centimeters. Record phase shift value.

4.5.3.3.10 Disconnect oscilloscope, ac current probe and 4.8 kc oscillator from test set.

4.5.3.4 Actuator controller test panel resistance test

4.5.3.4.1 Position FUNCTION switch as indicated below. Using resistance bridge, measure resistance between listed test points and COMMON. Set NORMAL/PULSE switch to PULSE position. Depress PUSH TO TEST switch as indicated.

PUSH TO TEST Switch Position	FUNCTION Switch Position	Test Point
Normal	1	LEFT UPPER RUDDER
Depressed	1	LEFT UPPER RUDDER
Normal	2	RIGHT UPPER RUDDER
Depressed	2	RIGHT UPPER RUDDER
Normal	3	LEFT LOWER RUDDER
Depressed	3	LEFT LOWER RUDDER
Normal	4	RIGHT LOWER RUDDER

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PUSH TO TEST Switch Position	FUNCTION Switch Position	Test Point
Depressed	4	RIGHT LOWER RUDDER
Normal	5	UP ELEVATOR
Depressed	5	UP ELEVATOR
Normal	6	DOWN ELEVATOR
Depressed	6	DOWN ELEVATOR

4.5.3.5 Actuator controller test panel functional test

4.5.3.5.1 Position all power switches to OFF. Connect 1000 ohm, 3-watt resistors between pins BB and HH, AA and HH, Z and HH of connector P1501. Resistors should remain in circuit until removal is specified.

4.5.3.5.2 Position POWER PANEL circuit breaker and ACTUATOR CONTROLLER TEST PANEL POWER switches to ON.

4.5.3.5.3 Using digital voltmeter, turn on +40 and -40 volt supplies and adjust supplies to 39.8 to 40.2 volts. Set NORMAL/PULSE switch to NORMAL.

4.5.3.5.4 With digital voltmeter, measure voltage between test points and COMMON jack for each control position as listed below:

Control	Position	Test Point
ROLL	CW	ROLL
YAW	RIGHT	YAW
PITCH	DOWN	PITCH
ROLL	CCW	ROLL
YAW	LEFT	YAW
PITCH	UP	PITCH

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